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Launching VPC Resources



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The screenshot displays the AWS Management Console interface for creating a new VPC. The left sidebar contains configuration options, and the right pane shows a visual preview of the network setup.

Configuration Options (Left):

- IPv4 CIDR block:** 10.0.0.0/16 (65,536 IPs). CIDR block size must be between /16 and /28.
- IPv6 CIDR block:** ☒ No IPv6 CIDR block, ☐ Amazon-provided IPv6 CIDR block.
- Tenancy:** Default.
- Number of Availability Zones (AZs):** 1, 2, 3. (2 is selected).
- Number of public subnets:** 0, 2. (2 is selected).
- Number of private subnets:** 0, 2, 4. (4 is selected).
- NAT gateways:** None, In 1 AZ, 1 per AZ. (None is selected).

Preview (Right):

- VPC:** nextwork-vpc
- Subnets (6):**
 - me-south-1a:** nextwork-subnet-public1-me-south-, nextwork-subnet-private1-me-, nextwork-subnet-private3-me-
 - me-south-1b:** nextwork-subnet-public2-me-south-, nextwork-subnet-private2-me-, nextwork-subnet-private4-me-
- Route tables (5):** nextwork-rtb-public, nextwork-rtb-private1-n, nextwork-rtb-private2-n, nextwork-rtb-private3-n, nextwork-rtb-private4-n



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Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a virtual network in AWS that lets you launch and control resources securely and isolate them from others.

How I used Amazon VPC in this project

We used Amazon VPC to host public and private EC2 instances, set up subnets, route tables, and security groups for secure networking.

One thing I didn't expect in this project was...

I didn't expect how easily private and public instances can communicate securely using security groups within the VPC.

This project took me...

This project took me an hour.



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Setting Up Direct VM Access

Directly accessing a virtual machine means logging into your EC2's operating system remotely, so you can run commands and manage it like a real computer.

SSH is a key method for directly accessing a VM

SSH stands for Secure Shell, a protocol that lets you securely access and manage a remote machine.

To enable direct access, I set up key pairs

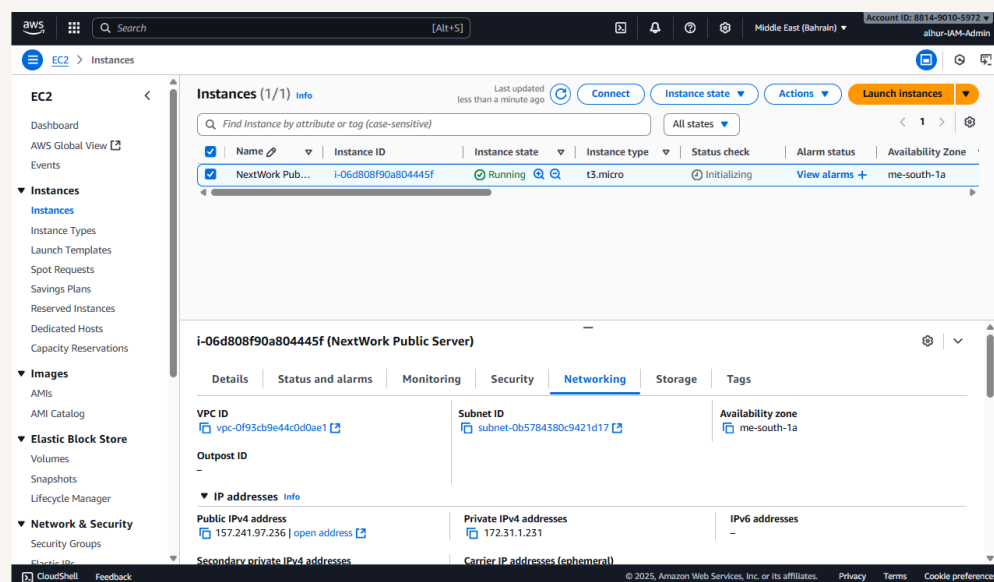
A key pair is a security credential with a public and private key that lets you securely access and manage your EC2 instance.

A private key is usually saved in .pem or .ppk format, and my private key was in the .pem format.



Launching a public server

I had to change my EC2 instance's networking settings by clicking on the edit network settings option in the network settings section when launching an instance.





Launching a private server

The private server uses a different security group to restrict access, allowing only trusted resources, unlike the public server which is open to the internet.

The new security group's source is the NextWork Public Security Group, allowing only resources in that group to access the private server.

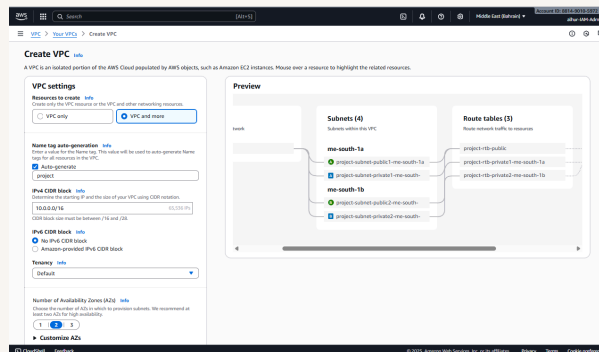
The screenshot displays the AWS Management Console interface for launching a new EC2 instance. The 'Launch an instance' page is active, showing various configuration options. The VPC is set to 'vpc-061036d8161543daf (NextWork VPC)' and the Subnet is 'subnet-0be77d9136f745b74 (NextWork Public Subnet)'. The 'Auto-assign public IP' option is disabled. Under the 'Firewall (security groups)' section, the 'Create security group' option is selected, and the security group name is 'NextWork Private Security Group'. The description is 'Security group for NextWork Private Subnet'. In the 'Inbound Security Group Rules' section, a rule is added with the following details: Type: ssh, Protocol: TCP, Port range: 22, Source type: Custom, Source: sg-049b905858c2b0b20, and Description: SSH for admin desktop. The Summary panel on the right provides a overview of the configuration: Number of instances: 1, Software image (AMI): Amazon Linux 2023 AMI 2023.8.2...read more, Virtual server type (instance type): t3.micro, Firewall (security group): New security group, and Storage (volumes): 1 volume(s) - 8 GiB. The 'Launch instance' button is highlighted in orange, and there are also 'Cancel' and 'Preview code' buttons.

Speeding up VPC creation

I used an alternative way to set up an Amazon VPC! This time, Instead of creating it from scratch , when i launched a vpc i selected VPC and more which allowed me to easily set it up from the same place.

A VPC resource map is a visual diagram showing all resources in your VPC—subnets, instances, gateways, and how they're connected.

Because each VPC is isolated, different VPCs can use the same IPv4 CIDR block without conflicting with each other.





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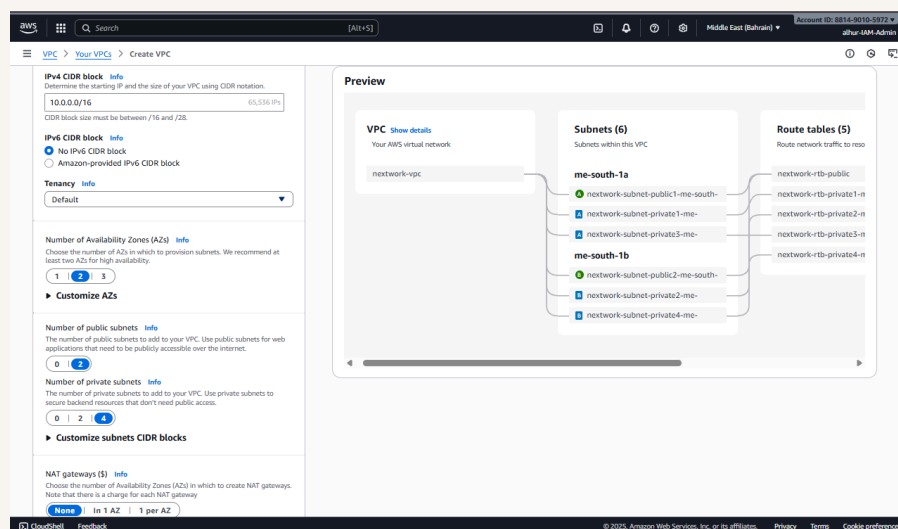
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Speeding up VPC creation

Tips for using the VPC resource map

When determining the number of public subnets in my VPC, I only had two options: zero and two.

NAT gateways let private subnet instances access the internet securely for updates while blocking incoming internet traffic.





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